

(e) Scatter diagrams and correlation		
2e.01	Know and apply vocabulary of correlation: positive, negative, zero, causation, association, interpolation and extrapolation.	Know that a dependent variable should be plotted on the 'y' axis.
2e.02	Describe and make comparisons of correlation by inspection: strong or weak.	e.g. Informal interpretation using scatter diagrams.
2e.03	<u>Know that correlation does not necessarily imply causation and multiple factors may interact.</u>	<u>Be aware of spurious correlation. e.g. car ownership and birth rate in a number of cities may show correlation as both variables are likely to be affected by population size of the cities.</u>
2e.04	Determine line of best fit by eye, by <u>drawing through a calculated double mean point $(\bar{x}, \bar{y})$</u> and by using the equation of the regression line.	The linear regression line of the form $y = a + bx$ Awareness of issues relating to interpolation and extrapolation, <u>and the interpretation of gradient and intercept are expected.</u> Non-linear models will not be tested.

Higher tier content

What students need to learn:		Guidance
(e) Scatter diagrams and correlation <i>continued</i>		
2e.05	Apply formula to determine Spearman's rank correlation coefficient. Values found using calculator functions will be permissible	Formula will be given in the formulae sheet. Tied ranks will not be tested. (<i>Scientific calculator functions are sufficient</i>).
2e.06	Interpret <u>calculated or</u> given Spearman's rank correlation coefficient in the context of the problem.	<u>Be aware that values range on a scale from -1 to +1. Know that values closer to these limits indicate 'stronger' correlation, but no formal interpretation of strength of correlation is expected.</u> <u>e.g. in comparing ranks given by two judges in a competition know that +1 means perfect agreement, -1 means complete opposite ranks, and 0 means no agreement between ranks given.</u>
2e.07	Interpret given Pearson's product moment correlation coefficient (PMCC) in the context of the problem.	Be aware that values range on a scale from -1 to +1. Know that values closer to these limits indicate 'stronger' linear correlation, but no formal interpretation of strength of correlation is expected. Know that +1 means perfect linear positive correlation, -1 means perfect linear negative correlation, and 0 means no linear correlation. The calculation of PMCC will not be required.
2e.08	Understand the distinction between Spearman's rank correlation coefficient and Pearson's product moment correlation coefficient (PMCC).	e.g. recognise the relative strengths of rank correlation and product moment correlation on a scatter graph. The PMCC measures the strength of linear correlation. The calculation of PMCC will not be required. e.g. if there is positive non-linear correlation both coefficients will be positive but Spearman's coefficient will be greater than PMCC.